



NII-UIT at VBS2026: Towards Effective Visual Question Answering for Interactive and Multimodal Video Retrieval

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Abstract. The Video Browser Showdown (VBS) provides a rigorous benchmark for evaluating interactive video retrieval systems under strict real-time constraints. While traditional tasks such as Known-Item Search (KIS) and Ad-hoc Video Search (AVS) remain central, the recent introduction of Visual Question Answering (VQA) has introduced new challenges. Participants are required to identify fleeting answer-bearing moments within long video sequences, often under ambiguous queries and limited time. Current systems face three main limitations: difficulties in accurately localizing these brief segments, the high verification effort demanded from users under time pressure, and limited capability to effectively integrate diverse modalities such as visual content and speech. To address these challenges, we present the NII-UIT system for VBS 2026, explicitly designed to advance VQA in interactive video retrieval. Our framework introduces an *Answer Span Prediction* module to highlight candidate temporal regions, a *Candidate Answer Suggestion* mechanism that aggregates multimodal cues to generate verifiable answer options, and a dedicated *In-Video Retrieval* component for frame-level evidence discovery. Complementary system optimizations and an enhanced user interface further improve interaction speed and reduce cognitive load. By shifting the user's role from open-ended searching to guided verification, our contributions are expected to improve efficiency and accuracy, offering a practical step toward more effective VQA systems in competitive evaluation scenarios.

Keywords: Interactive video search · Video Retrieval System · Visual Question Answering

1 Introduction

The Video Browser Showdown (VBS) [8] is an annual international competition that has served as the leading benchmark for interactive video retrieval technolo-

gies. Organized as a live session at the International Conference on Multimedia Modeling (MMM), the competition provides a unique testbed where retrieval systems are evaluated under time pressure, uncertain conditions, and direct human interaction. By combining the rigor of benchmarking with the dynamics of a game-like challenge, VBS has become both a showcase of cutting-edge techniques and a driver of innovation in multimedia search. To ensure diversity and realism, the competition employs large-scale and heterogeneous video datasets such as V3C [9], Marine Video Kit (MVK) [12], and LapGynLHE [6], which cover everyday, scientific, and medical domains, respectively.

Beyond its traditional tasks, such as Known-Item Search (KIS) and Ad-hoc Video Search (AVS), VBS has progressively introduced more complex formats to push the limits of current systems. Question Answering (QA) [3], first appearing in VBS 2024 [10], requires participants to locate precise answers to natural-language questions within long video sequences, answers that may appear only briefly, demanding fine-grained temporal navigation. With VBS 2026, QA tasks are more frequent and often combined with extended forms of KIS, such as KIS with text (KIST) and KIS with chat (KISC). In KISC, players may query the organizer for hints, but the responses are partial or indirect, requiring inference and reasoning to identify the correct segment. KIST, on the other hand, challenges participants to comprehend, remember, and link together multiple textual descriptions across time, something most retrieval systems are not designed to handle, as they typically process only isolated queries.

Despite these advances, several gaps remain that limit the effectiveness of current systems in handling VQA tasks at VBS. First, existing pipelines [11] often fail to localize short-lived evidence within long videos, forcing participants into inefficient manual browsing. Second, even when relevant segments are identified, users must still spend significant effort verifying the exact answer, a process that is prone to error under strict time constraints. Finally, conventional retrieval systems are typically optimized for video-level search, leaving a gap between coarse retrieval and the fine-grained, frame-level localization demanded by VQA.

To overcome these limitations, we propose a set of complementary improvements specifically tailored for VQA in interactive video retrieval. At the core, an *Answer Span Prediction* module leverages vision-language models to highlight candidate temporal segments, reducing the search space and guiding users toward answer-bearing regions. Building on this, a *Candidate Answer Suggestion* mechanism aggregates captions and transcripts from predicted hotspots to generate multiple plausible answer options, enabling rapid verification rather than exhaustive exploration. To close the gap between video-level retrieval and fine-grained evidence discovery, we further introduce an *In-Video Retrieval* module that performs multimodal search directly over keyframes, enhancing temporal precision. Finally, we accelerate the overall pipeline and redesign the user interface to provide synchronized multimodal views, interactive timelines, and candidate answer panels, thereby streamlining the verification process and improving usability. Together, these contributions advance the state of VQA in VBS by shifting the user’s role from manual searching to guided decision-making,

ultimately improving both efficiency and effectiveness under competition constraints.

2 Previous NII-UIT System Overview

The NII-UIT system [2], which won the overall VBS 2025 competition, was built as a multimodal retrieval framework organized into offline and online stages. In the offline stage, videos were segmented into representative keyframes, which were embedded into a vector database along with semantic features extracted by deep models. An object detection module further indexed visual entities, providing structured metadata for filtering during search.

In the online stage, the system supported multiple query modalities. Textual and visual inputs were projected into a shared embedding space via vision-language models for cross-modal retrieval. To improve robustness, a query expansion module [5] generated paraphrased variations, while a visual query generation component converted textual descriptions into synthetic images [7], extending the representational scope of queries. Object filtering and temporal search modules were also incorporated, enabling refinement by entity constraints and navigation across consecutive query contexts.

The system was delivered through a web-based interface designed for both novice and expert users. It featured simplified interaction for accessibility, while advanced controls allowed experts to adjust model weights, reformulate queries, and integrate multimodal evidence. Together, these modules provided a flexible and efficient architecture that balanced usability and accuracy in the VBS 2025 competition.

3 Improvements

3.1 Better Vision Language Models

In VBS 2025, the NII-UIT system adopted CLIP [1] and BEIT-3 [13] as its primary vision-language encoders. CLIP, trained on large-scale image-text pairs [14], provided a strong baseline for cross-modal alignment. At the same time, BEIT-3 contributed enhanced multimodal understanding through joint pretraining across image, text, and vision-language tasks. However, despite their effectiveness, CLIP’s reliance on noisy web-scale data sometimes led to unstable retrieval performance, especially in fine-grained scenarios such as locating short-lived video segments or interpreting nuanced textual descriptions.

For VBS 2026, we replaced CLIP with SigLIP [15], a recent extension designed to improve semantic consistency and robustness. SigLIP refines the alignment between vision and language embeddings by incorporating signal-level regularization during training, which reduces semantic drift and improves generalization across domains. Compared to CLIP, SigLIP demonstrates more reliable grounding of textual queries to relevant video content, particularly when queries involve subtle attributes or complex compositions.

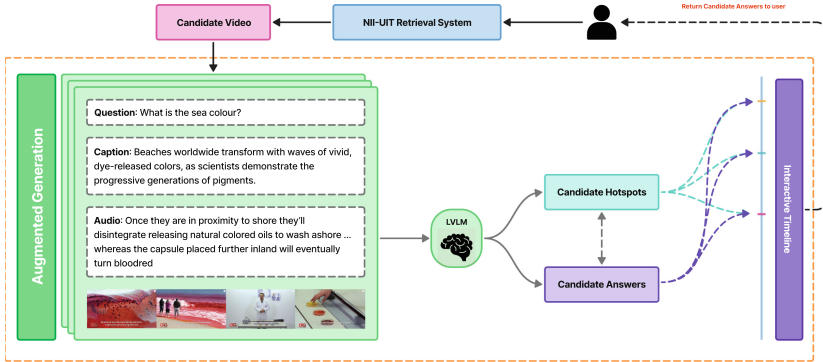


Fig. 1. Overview of the proposed VQA pipeline. The system fuses multimodal cues – captions, audio transcripts, and keyframes – through a Large Vision-Language Model (LVLM) to predict candidate hotspots and suggest plausible answers. Both are visualized on an interactive timeline for rapid verification and user selection.

3.2 Visual Question Answering

Answer Span Prediction. In QA tasks, the evidence supporting the correct answer often appears only fleetingly within several minutes of video content, which makes manual browsing both inefficient and error-prone. Current retrieval pipelines typically lack mechanisms for automatically highlighting the temporal segments that are most likely to contain the required information.

We therefore propose an Answer Span Prediction module that integrates a Large Vision-Language Model (LVLM) with dense video captions and automatically generated transcripts. The central idea is not to rank every frame by similarity, but rather to identify those temporal segments that contain sufficient semantic cues to answer the user’s question (Fig. 1), and to propose them as candidates for inspection. The process begins by segmenting the video into short temporal units enriched with fine-grained captions generated by vision–language models. The user’s natural-language question is then processed by an LVLM named NVILA [4], which aligns its semantic representation with the caption timeline to filter out segments unlikely to contain relevant evidence. The outcome is a set of candidate “hotspots” that are presented to the user as the most plausible answer-bearing regions.

Candidate Answer Suggestion. While Answer Span Prediction reduces the search space, users still need to inspect hotspots to find the exact answer. This inspection step can be inefficient under the strict time limits of VBS. To further accelerate the process, we propose a *Candidate Answer Suggestion* module that directly exploits captions and speech transcripts extracted from the hotspots.

For each hotspot, the system collects short textual cues from automatically generated visual captions and spoken dialogue transcripts. These textual cues are subsequently aligned with visual evidence from the corresponding frames to

ensure that the proposed answers are both semantically consistent and visually grounded. The combined information is then processed together with the user’s question to derive a small set of n candidate answers. Rather than attempting to produce a single free-form response, the system generates multiple plausible alternatives, reflecting different but contextually valid interpretations of the available evidence. This candidate set is shown to users for quick verification and selection, minimizing errors from noisy captions or transcripts and improving efficiency by turning open-ended searching into focused verification.

3.3 In-Video Retrieval

In many VQA tasks, users may successfully locate the correct video but still struggle to identify the precise frames that contain the answer-bearing evidence. This gap between video-level retrieval and frame-level localization often leads to wasted time, as participants must manually scan through long sequences under strict competition deadlines.

To address this challenge, we introduce a dedicated *Search in Video* module designed for fine-grained temporal navigation. Once a video has been identified as a candidate, it proceeds with a multimodal search over the extracted keyframes from that video. Specifically, the multimodal search integrates text-based retrieval, image-based similarity search, object-level filtering, on-screen text recognition, and spoken content analysis. The resulting candidate frames are presented along the video timeline, ranked by their similarity to the user’s query, so that participants can quickly focus on the segments most likely to contain the correct answer.

3.4 System Acceleration and Optimized User Interface

System Acceleration. Compared with the previous year, the retrieval pipeline of the NII-UIT system was substantially optimized to improve efficiency under strict time constraints. Several key changes were introduced to accelerate query processing. First, extensive profiling was conducted to identify and eliminate redundant computations, resulting in a cleaner and more efficient retrieval pipeline. Second, unnecessary code branches and duplicate feature extraction steps were removed, reducing both memory usage and execution time. Third, caching mechanisms were introduced for repeated or paraphrased queries, allowing the system to return results without redundant computation immediately.

Optimized User Interface. To support the increasing complexity of VQA tasks, the user interface was redesigned with a focus on clarity, speed, and multimodal interaction. Key improvements include an interactive timeline that highlights candidate hotspots identified by the system, allowing users to quickly navigate to frames most likely to contain relevant answers. For each hotspot, preview keyframes are displayed directly on the timeline, enabling rapid visual inspection without full playback.

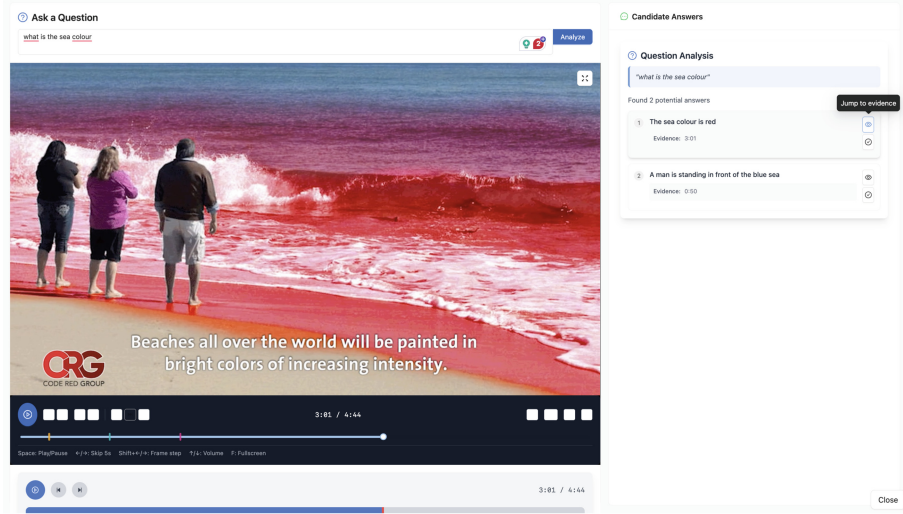


Fig. 2. Optimized VQA interface showing timeline hotspots and answer options with linked evidence, helping users verify answers via visual, transcript, and multimodal cues efficiently.

The UI also integrates synchronized transcript and caption panels, where spoken dialogue and automatically generated descriptions are aligned with video playback. This helps participants cross-check textual evidence with corresponding frames. In addition, a candidate answer panel was introduced to present multiple plausible answers derived from multimodal analysis, shifting the user’s role from open-ended searching to quick verification (see Fig. 2).

4 Conclusion

In conclusion, our VBS 2026 system advances interactive video retrieval with a dedicated focus on Visual Question Answering. By introducing answer span prediction and candidate answer suggestion, the system effectively narrows the search space and accelerates verification. A complementary in-video retrieval module further improves frame-level localization, while system optimizations and an enhanced user interface streamline interaction under strict time constraints. Collectively, these enhancements make VQA a more guided, efficient, and accurate process suited to competitive evaluation settings.

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